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CS 499 Computer Science Capstone

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CS 499 Professional Self-Assessment

Proceeding through and completing the computer science program at SNHU with the totality of my efforts being dedicated to finalizing my capstone ePortfolio project has required my full effort and initiative to transform my knowledge gained from the course work into a completed body of work that has the potential to be extended into a fully functioning web application. My only artifact for this capstone project was an application that I named, InvenX, which was initially project from CS-360 and was a Java based Android app inventory management project designed with a local SQLite database. InvenX was derived from that Android app into what is now a full MEAN stack web application with secure REST endpoints, JWT-based authentication, sanitized inputs, robust client-side filtering and search, and a custom Merge Sort function. The application is predominantly written in TypeScript, JavaScript and HTML. Transitioning the languages of the original Java and SQLite mobile app to a web application running MongoDB, Mongoose and Angular for the front end enhanced my ability to build full stack web application, and how to secure the applications endpoints, as the first Android application lacked any significant security or sanitation. Thanks to the process of this design, and the learning and testing needed to transition to different languages, I feel like I am able to at the very least display knowledge of systems, and architecture of design processes when building an application.

I wanted to have a strong focus on cybersecurity, as this is my penultimate goal for a potential career choice, or possibly a general IT role initially, in which I highlight my knowledge of troubleshooting, and security rather than purely software development. During the program at SNHU, I learned to practice communication with mock stakeholders through documentation, and how to use the software development life cycle to properly translate technical decisions into a working application. Throughout the capstone project, through the usage of milestones, I updated my InvenX app in a regimented, iterative process and communicated my thought process and design choices. The capstone, due to its open ended design in education, allowed me to have more freedom of design choices in how I wanted to implement the project.

One of the most difficult aspects of the design of the project was working through the section of data structures and algorithms, which needed a design of an advanced algorithm that met standards of Big O notation for time and space complexity. Developing the advanced algorithm first required me to move from theory to implementation of different design ideas,

which began with simply initiating a built in `sort()` function from JavaScript and once completed enough of the project to have functioning sort, performing required research and then creating and integrating a custom Merge Sort function that supported dynamic fields and ascending or descending order. On research, the built in `sort()` for JavaScript had a time complexity of best case $O(N \log N)$ and worst case of $O(N^2)$, which I felt, while being adequate has poor, worst case scenario and did not meet the goals of the advanced algorithm requirements. Also, the time complexity, from what I read, is based on the version of JavaScript so the best case may be worse than the provided $O(N \log N)$. Instead of sticking to a built in `sort()`, I designed a stable merge sort function with a best and worst case of $O(N \log N)$, while keeping my goals to have support for dynamic fields and ascending or descending ordering of fields. Other design choices for sorting were my client-side filtering, which allowed for text search, range filtering for quantity, price, and reorder threshold, and category selection of items via checkboxes. While not algorithmic in nature, I felt that this design choice expanded upon the original Android app, displayed my ability and brought the application a feel of having realistic user organization. I based the design choices off of the website design of PC Part Picker for slider bars and checkboxes, and while I am happy with the implementation, I would have liked to adjust the slider bars to have dual ended usage to create a range, but as this not standard in HTML or JavaScript, I would have had to design the function myself, and did not have the time for implementation. But, in saying all of this, I want to continue with the challenge of implementing this design once completed with this capstone course. All this pertains to my target roles in cybersecurity or IT through my improved understanding of algorithmic design and increased fluency in Big O notation, and the actual design and implementation of the algorithm.

My software engineering and database skills improved significantly as I developed an entirely different backend with Express and MongoDB, as compared to an SQLite local database, and I learned more about testing endpoints of the API via Postman. My goal throughout the process was to create secure REST endpoints with enforced validation in Mongoose models. I wrote standard CRUD operations that were tested and provided errors and linked this implementation to the Angular UI so that it could be displayed properly and reliably despite user's improper input of fields. In order to maintain my security posture for development, I added route guards, form validation, meaningful error messages that did not expose intricate information about the background processes of the application that could be utilized by a potential threat actor, and implemented a consistent UI design throughout the application. Other security design choices in InvenX that were not handled in the original application were how I handled search input sanitation, and protected the logout and URL flow to prevent leaking of information when a user was not logged into the application. I plan to take this same discipline and security first mindset into an IT or security role with the intention to define boundaries of the application, and reduce surface of attack for threat actors.

I have mentioned this throughout my paper, but the mainstay of what I was trying to achieve was to display my knowledge of security in my work. One of my main focuses, and once

development and testing of the initial database was completed, I implemented JWT authentication and kept tokens on the client, and then utilized the tokens, or lack thereof, to validate requests to the server. My example being, inability to reach the inventory page without being logged into the application, any attempt will transition the user back to the initial log in page. As I stated previously, I sanitized inputs on the client side and used the backend as the ultimate authority in this decision process. With my goals and interests being predominantly pointed at cybersecurity and general IT, as I intend to continue this journey through obtaining different certifications, I enjoyed the time I expended, effort I put forth, and knowledge I gained while learning the ins and outs of hardening systems, and making defensible choices that reduce risk and attack surfaces. The capstone project gave me confidence that I can bring to a security environment, as I now have a fundamental understanding of design and development choices and operations in which web based applications are created.

The organization and setup of my ePortfolio provides all the information to display, not only my thought processes throughout my design of my only artifact, but also, links to the files in which one could evaluate my code practices. The original Android inventory app is a simple design, which houses single device storage via the Java language, and local SQLite, while my enhanced artifact covers software design and engineering through writing the application in another language, algorithms and data structures through my created merge sort algorithm, and databases through transitioning to a non-relational MongoDB. The finalized, temporarily as I have plans to extend functionality, InvenX app displays my ability to write a full-stack web app throughout the entire design process with secure APIs, JWT auth, MongoDB, and a front end that is modern, and consistent. My code review video informs reviewers of my design choices and reasoning for these choices, as most of the design ideas focused on the implementation of improved security practices. My downloadable and viewable milestone narratives on the GitHub page displays my thought process throughout the artifact design. Since I wanted to allow the user to view the milestones without downloading, I converted the Word narratives to embedded PDFs for reading on the page.

As I consider my future career motivations and move toward my goal of obtaining a career in the cybersecurity field, I plan to use this ePortfolio as evidence that I can build, debug, and secure web applications end to end with the knowledge that I can extend this concept to further systems of design. My work shows my thought processes of how I evaluate potential risks, and back my choices with working code and modular code with secure configurations. My goal is to gain an entry level position in the cybersecurity field or in an end user operation general IT role that I can use to further grow through ongoing practice, reading and certifications into a security focused position, like a cybersecurity specialist or penetration tester. Based on my ePortfolio and artifact, it is clear that I can contribute at once to supporting and improving systems, and continue building upon tools and documentation that make those systems safer for end users.